Chapter 6—DNA: How The Molecules of Heredity Carries, Replicates, and Recombines Information

Fill in the Blank

1. The full chemical name of DNA is __________________.
   Ans: deoxyribonucleic acid
   Difficulty: 2

2. The protein, ________, is needed to lay down a segment of RNA complementary to the DNA before replication can begin.
   Ans: primase
   Difficulty: 2

3. An origin of replication generally has _______ (how many?) replication forks.
   Ans: two
   Difficulty: 2

4. The discontinuous lagging strand on a replication fork consists of _________ fragments that are later ligated together to form one strand of DNA.
   Ans: Okazaki
   Difficulty: 2

5. The *E. coli* chromosome contains _____ (how many?) origin(s) of replication?
   Ans: one
   Difficulty: 3

6. A prokaryote lacks a __________ membrane.
   Ans: nuclear
   Difficulty: 2

7. Recombination is probably always initiated by _______ of DNA.
   Ans: nicking
   Difficulty: 3

8. The _________ strand is synthesized continuously during DNA replication.
   Ans: leading
   Difficulty: 2

9. A _____________ enzyme recognizes a specific DNA sequence and is able to cleave the DNA generating either “blunt” or “sticky” ends of the DNA.
   Ans: restriction
   Difficulty: 1

10. A virus that infects bacterial cells is called a ______________.
Ans: bacteriophage
Difficulty: 2

11. The genetic material of some viruses such as ϕX174 and M13 is _____________.
Ans: single-stranded DNA
Difficulty: 3

12. The genetic material of the viruses that cause polio and AIDS is ________.
Ans: RNA
Difficulty: 3

13. Topoisomerase _________ supercoils by nicking the DNA.
Ans: relaxes
Difficulty: 2

14. In *E. coli*, recombination begins at _____ sites where recBCD nicks and displaces a strand.
Ans: chi
Difficulty: 2

<u>Multiple Choice</u>

15. Which of the following is not true of DNA?
A) It is acidic.
B) It contains deoxyribose.
C) It is found in cell nuclei.
D) It contains phosphate.
E) It contains proteins.
Ans: E
Difficulty: 2

16. The molecule of heredity is:
A) RNA.
B) DNA.
C) protein.
D) carbohydrate.
E) none of the above
Ans: B
Difficulty: 1

17. The four subunits which compose DNA are called:
A) phosphodiesters.
B) proteins.
C) nucleotides.
D) nucleosides.
E) polymers.
Ans: C
Difficulty: 2

18. DNA is localized mainly in the:
A) cell membrane.
B) endoplasmic reticulum.
C) vacuoles.
D) chromosomes.
E) none of the above
Ans: D
Difficulty: 2

19. Each nucleotide of DNA is made up of:
A) a deoxyribose sugar.
B) a nitrogenous base.
C) a phosphate.
D) a and b only.
E) all of the above
Ans: E
Difficulty: 2

20. The polarity of DNA synthesis is:
A) $5' \rightarrow 3'$.
B) $3' \rightarrow 5'$.
C) $5' \rightarrow 2'$.
D) $2' \rightarrow 5'$.
E) none of the above
Ans: A
Difficulty: 1

21. In the Hershey and Chase experiment designed to determine the molecule of heredity, what was radiolabeled with ^{35}S?
A) protein
B) DNA
C) RNA
D) rRNA
E) none of the above
Ans: A
Difficulty: 2

22. The ratio of _____ is 1:1.
A) guanine to adenine
B) adenine to thymine
C) cytosine to adenine
D) uracil to cytosine
E) none of the above
Ans: B
Difficulty: 2

23. X-ray data showed that the spacing between repeating units along the axis of the DNA helix is:
A) 2.0 angstroms.
B) 3.4 angstroms.
C) 20 angstroms.
D) 34 angstroms.
E) none of the above
Ans: B
Difficulty: 3

24. X-ray data showed that the DNA helix undergoes one complete turn every:
A) 2.0 angstroms.
B) 3.4 angstroms.
C) 20 angstroms.
D) 34 angstroms.
E) none of the above
Ans: D
Difficulty: 3

25. ________ bonds are responsible for the chemical affinity between A and T (or G and C) nucleotides.
A) Ionic
B) Covalent
C) Hydrogen
D) Electro-ionic
E) none of the above
Ans: C
Difficulty: 2

26. _______ -form DNA spirals to the right and is the major form of naturally occurring DNA molecules.
A) A
B) B
C) D
D) Y
E) Z
Ans: B
Difficulty: 2

27. The nucleotide that is present in RNA but not DNA is:
A) thymine.
B) uracil.
C) adenine.
D) cytosine.
E) guanosine.
Ans: B
Difficulty: 1

28. DNA replication occurs through a _________process.
A) conservative
B) semiconservative
C) dispersive
D) transferal
E) none of the above
Ans: B
Difficulty: 2

29. During complementary base pairing, enzymes join the base's nucleotide to the preceding nucleotide by a __________bond.
A) hydrogen
B) ionic
C) phosphodiester
D) electrostatic
E) none of the above
Ans: C
Difficulty: 2

30. During early interphase the state of the DNA can be described as:
A) a single continuous linear double helix.
B) a double helix replicated semiconservatively.
C) a double helix replicated conservatively.
D) single-stranded DNA.
E) none of the above
Ans: A
Difficulty: 4

31. During the S phase of interphase, the state of the DNA can be described as:
A) a single continuous linear double helix.
B) a double helix replicated semiconservatively.
C) a double helix replicated conservatively.
D) a triple helix replicated semiconservatively.
E) none of the above
Ans: B
Difficulty: 4

32. The step in DNA replication in which the replication proteins open up the double helix and prepare for complementary base pairing is called:
A) initiation.
B) elongation.
C) termination.
D) translation.
E) translocation.
Ans: A
Difficulty: 2

33. The step in DNA replication in which the proteins connect the correct sequence of nucleotides into a continuous new strand is called:
A) initiation.
B) elongation.
C) termination.
D) translation.
E) translocation.
Ans: B
Difficulty: 2

34. The step in DNA replication in which two replication forks moving in opposite directions may meet is called:
A) initiation.
B) elongation.
C) termination.
D) translation.
E) translocation.
Ans: C
Difficulty: 2

35. The group of enzymes able to relax supercoils in DNA is called:
A) primases.
B) helicases.
C) topoisomerases.
D) telomeres.
E) ligases.
Ans: C
Difficulty: 2

36. How many replication forks depart from an origin of replication?
A) one
B) two
C) three
D) four
E) none of the above
Ans: B
Difficulty: 2

37. The protein that progressively unwinds DNA ahead of each replication fork is called:
A) primase.
B) helicase.
C) topoisomerase.
D) telomerase.
E) ligase.
Ans: B
Difficulty: 2

38. In eukaryotic cells, replication proceeds from ____ origin(s) of replication.
A) no
B) one
C) two
D) several
E) many
Ans: E
Difficulty: 3

39. Eukaryotic chromosomes have evolved special structures at the ends of chromosomes to ensure the replication of the two ends of linear chromosomes. These structures are called:
A) methylases.
B) capping proteins.
C) ligases.
D) telomeres.
E) single-stranded biding proteins.
Ans: D
Difficulty: 2

40. Which of the following is <u>not</u> involved in ensuring the accuracy of a cell's genetic information?
A) redundancy
B) repair enzymes
C) precision of replication machinery
D) DNA polymerase proofreading mechanism
E) restriction endonucleases
Ans: E
Difficulty: 3

41. The process of __________ is extremely important in generating genetic diversity.
A) translation
B) transcription
C) recombination
D) transformation
E) none of the above
Ans: C
Difficulty: 2

42. Any deviation from the expected 2:2 segregation of parental alleles that results from recombination is known as:
A) allelic exchange.
B) gene conversion.
C) crossing over.
D) recombination.
E) DNA replication.
Ans: B
Difficulty: 3

43. Recombination occurs during meiotic ________.
A) anaphase
B) interphase
C) prophase
D) metaphase
E) none of the above
Ans: C
Difficulty: 3

44. Recombination involves the breakage and reunion of DNA molecules from:
A) homologous nonsister chromatids.
B) homologous sister chromatids.
C) heterologous nonsister chromatids.
D) heterologous sister chromatids.
E) none of the above
Ans: A
Difficulty: 2

45. The nicking of DNA that initiates recombination during mitosis may be due to all but:
A) instruction from normal cell-cycle program.
B) X-rays.
C) chemical damage.
D) physical damage.
E) radiation.
Ans: A
Difficulty: 3

46. What radiolabeled substance did Hershey and Chase use to label the protein component of the bacteriophage in their study to determine whether protein or DNA was necessary for phage production?
A) nitrogen
B) carbon
C) sulfur
D) phosphorous
E) iodide
Ans: C
Difficulty: 2

47. If 35% of the bases in a region of the mouse genome are cytosine, what percentage in that region are adenine?
A) 15%
B) 20%
C) 30%
D) 35%
E) none of the above
Ans: A
Difficulty: 3

48. The complementary sequence of 5' AATTCGCTTA 3' is:
A) 5' AATTCGCTTA 3'.
B) 3' AATTCGCTTA 5'.
C) 5' TAACGCTTAA 3'.
D) 5' TAAGCGAATT 3'.
E) 3' TAAGCGAATT 5'.
Ans: D
Difficulty: 2

49. Two strains of *S. cerevisae* (yeast) are crossed. One has the genotype *ABC* and the other *abc*. Five sets of the resultant tetrads are noted below. In which set did a gene conversion event occur?
A) *abc*, *aBc*, *AbC*, *aBC*
B) *abc*, *abc*, *ABC*, *ABC*
C) *aBc*, *aBc*, *AbC*, *AbC*
D) *abC*, *abc*, *ABc*, *ABC*
E) *Abc*, *Abc*, *aBC*, *aBC*
Ans: A
Difficulty: 4

50. Without ___________ regions on the DNA during recombination, gene conversion could not occur.
A) homoduplex
B) heteroduplex
C) homotriplex
D) heterotriplex
E) none of the above
Ans: B
Difficulty: 2

51. Occasionally, a loss of function mutation may occur in the telomerase enzyme in a cell. What is likely to be the result of this mutation on the DNA in the cell over the course of several rounds of mitosis?
A) Chromosome length will gradually increase.
B) Chromosome length will gradually decrease.
C) Chromosome length will stay constant.
D) Chromosomes will fail to dissociate after replication.
E) None of the above
Ans: B
Difficulty: 3

52. Bacterial DNA is resistant to degradation by its own restriction enzymes through the protection of:
A) methylases.
B) recombinases.
C) topoisomerases.
D) ligases.
E) primases.
Ans: A
Difficulty: 2

53. Hershey and Chase relied on _________ to physically separate the infected bacterial cells from the phage ghosts.
A) radioactivity
B) gel filtration
C) ion exchange
D) centrifugation
E) none of the above
Ans: D
Difficulty: 3

54. Meselson and Stahl relied on equilibrium density gradient centrifugation in a ________ solution to resolve the DNA containing ^{14}N from the DNA containing ^{15}N.
A) radiolabeled phosphate
B) calcium chloride
C) radiolabeled nitrogen
D) sodium acetate
E) cesium chloride
Ans: E
Difficulty: 3

Matching

Match the scientist with their contribution to understanding more about DNA
a. early recombination models
b. transformation
c. semiconservative replication
d. x-ray structure of DNA
e. revised recombination models
f. basic structure of DNA
g. DNA = molecule of heredity
h. discovery of "nuclein"
i. nucleotide ratios in DNA
j. famous phage blender experiment

55. ___ Watson and Crick
Ans: f
Difficulty: 3

56. ___ Chargaff
Ans: i
Difficulty: 3

57. ___ Franklin and Wilkins
Ans: d
Difficulty: 3

58. ___ Griffith
Ans: b
Difficulty: 3

59. ___ Hershey and Chase
Ans: j
Difficulty: 3

60. ___ Miescher
Ans: h
Difficulty: 3

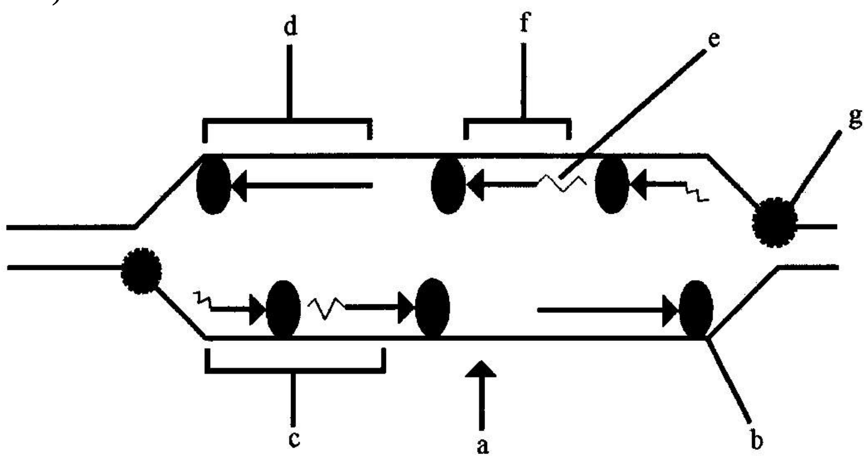
61. ___ Meselson and Stahl
Ans: c
Difficulty: 3

62. ___ Holliday
Ans: a
Difficulty: 3

63. ___ Meselson and Radding
Ans: e
Difficulty: 3

64. ___ Avery, MacLeod and McCarty
Ans: g
Difficulty: 3

Diagram Matching (Match the region or item on the diagram with the correct term from the list below)



65. ___ lagging strand
Ans: c
Difficulty: 2

66. ___ leading strand
Ans: d
Difficulty: 2

67. ___ origin
Ans: a
Difficulty: 2

68. ___ RNA primer
Ans: e
Difficulty: 2

69. ___ Okazaki fragment
Ans: f
Difficulty: 2

70. ___ helicase
Ans: g
Difficulty: 2

71. ___ polymerase
Ans: b
Difficulty: 2

True or False

72. A phosphodiester bond joins one nucleotide together in the DNA polymer.
Ans: True
Difficulty: 1

73. Prokaryotes have a circular chromosome surrounded by a nuclear membrane.
Ans: False
Difficulty: 1

74. Transformation in bacteria results from the uptake of foreign DNA.
Ans: True
Difficulty: 1

75. DNA is highly negatively charged throughout the molecule and therefore has no polarity.
Ans: False
Difficulty: 1

76. A region of DNA 100 bp in length has the potential to be represented by 4^{100} unique sequences.
Ans: True
Difficulty: 1

77. Viruses use only DNA as their genetic material.
Ans: False
Difficulty: 1

78. Restriction enzymes are molecular weapons which cut DNA that bacteria use in their fight to protect themselves from DNA viruses.
Ans: True
Difficulty: 1

79. Human DNA is replicated in a conservative manner.
Ans: False
Difficulty: 1

80. In DNA replication, new DNA is produced in a continuous bi-directional fashion.
Ans: False
Difficulty: 1

81. DNA topoisomerase supercoils DNA to get it out of the way of DNA polymerase.
Ans: False
Difficulty: 2

82. Accuracy of replication is enhanced by the redundancy of DNA, enzymatic repair of damaged DNA and the remarkable precision of the cellular replication machinery.
Ans: True
Difficulty: 1

83. A 2:2 segregation of parental alleles is known as gene conversion.
Ans: False
Difficulty: 2

84. Recombination events are limited to only "hot spots" along a chromosome.
Ans: False
Difficulty: 1

85. Gene conversion in which a small segment of information from one homologous chromosome transfers to the other can give rise to an unequal yield of two different alleles.
Ans: True
Difficulty: 2

86. Alternative resolutions of the Holliday intermediate are responsible for whether or not crossing over or gene conversion occurs.
Ans: True
Difficulty: 2

Short Answer

87. Draw the complementary strand to the one shown below. Be sure to note polarity.
5' CATAGCCTTA 3'
Ans: 5' TAAGGCTATG 3'
Difficulty: 2

88. The enzyme *Hin*dIII cleaves DNA between the two adenine bases in the specific sequence seen below. Indicate where the cuts would occur on each strand of the DNA below. Indicate what type of DNA ends result from this cleavage.
 5' AAGCTT 3'
 3' TTCGAA 5'
Ans: 5'A AGCT T 3'
3'TTCGA A 5'
sticky or cohesive ends
Difficulty: 2

89. Briefly describe the role of an initiator protein in DNA replication.
Ans: The initiator protein is the first protein to bind to the origin of replication and attracts helicase which is able to unwind the region surrounding the origin—the first step in initiation of DNA replication.
Difficulty: 3

90. During the DNA replication process, RNA primers are removed and replaced with DNA. Briefly state why it is important that this RNA be replaced with DNA.
Ans: RNA is more unstable than DNA (it may be easily degraded). DNA must also be present so that the next round of replication may proceed.
Difficulty: 4

91. Why are DNA molecules considered redundant?
Ans: Either strand of the double helix can specify the sequence of the other.
Difficulty: 2

92. The Holliday model of recombination has been modified and is now called the *consensus model*, which is now consistent with current research. What are the five properties of recombination, as they are now understood?
Ans: 1. homologs physically break, exchange parts, and rejoin
2. breakage and repair create reciprocal products of recombination
3. recombination events can occur anywhere along the DNA molecule
4. the exchange is precise, there is no gain or loss of nucleotides
5. gene conversion can give rise to an unequal yield of two different alleles
Difficulty: 4

93. What are the eight steps of recombination (crossing over)?
Ans: 1. the DNA strand is nicked
2. helicase unwinds the DNA and whisker displacement occurs
3. first strand invasion occurs
4. second strand invasion occurs
5. the D-loop is degraded and the gaps are repaired and ends are ligated
6. branch migration lengthens the heteroduplex region
7. chromatid rotation forms a Holliday intermediate
8. endonucleases cut the strands and the chromatids are resolved.
Difficulty: 4

94. How is it possible for an individual to be XX male or XY female?
Ans: An illegitimate recombination event can take place between the X and Y chromosomes. The result is a Y chromosome that lacks the *SRY* region or an X that has gained it.
Difficulty: 3

95. How does mismatch repair lead to gene conversion?
Ans: Repair can proceed in two ways, correction of the incorrect strand or change of the correct strand to the incorrect sequence.
Difficulty: 3

96. What is a heteroduplex region?
Ans: The segment that is located between the annealed break points of recombination is called heteroduplex because one strand is maternal and one is paternal and mismatch may be present.
Difficulty: 3

Experimental Design and Interpretation of Data

97. An illegitimate recombination event can take place between the human X and Y chromosomes. The result may be a sperm with a Y chromosome which lacks *SRY* (the sex-determining region). Fertilization of an egg by this particular sperm would yield what type of individual. State both genotype and phenotype.
Ans: XY individual that develops as a female.
Difficulty: 3

98. A circular 5.0 kb piece of DNA is cut with a restriction enzyme, *Eco*RI, which cleaves the DNA twice. One fragment is 3.0 kb, while the other is 2.0 kb. When the same 5.0 kb piece of DNA is cleaved with *Hin*dIII, only one cut is made. Briefly describe how you would determine whether the *Hin*dIII site resides in the 3.0 kb *Eco*RI fragment or in the 2.0 kb *Eco*RI fragment. (Assume that the *Hin*dIII site is at least 500 bp away from each of the *Eco*RI sites.)
Ans: A double digest with both *Eco*RI and *Hin*dIII would determine which *Eco*RI fragment is further cleaved with *Hin*dIII.
Difficulty: 4

99. Griffith found that smooth (S) forms of *S. pneumoniae* have a polysaccharide capsule and rough (R) forms do not. Only S forms cause infection. Briefly describe how Griffith demonstrated transformation using live R form and heat killed S form bacteria.
Ans: S^{HK} does not yield infection R^{live} does not yield infection S^{HK}+R^{live} = infection, R has been transformed by S.
Difficulty: 2

100. When Meselson and Stahl performed the experiment that showed that replication is a semiconservative process, they utilized *E. coli*, and various isotopes of nitrogen (^{15}N and ^{14}N). Explain briefly what their results <u>would have been</u> if DNA replicated conservatively.
Ans: Following centrifugation, the first generation of replication would yield 2 bands - ^{15}N and ^{14}N (no hybrid). The second generation would again result in the same pattern with no hybrid pattern ever revealed.
Difficulty: 4

101. You are a researcher at a new Biotech company. You have been asked to devise a scheme to use bacteria to produce protein X, which has been found important in cancer treatment in humans. Protein X is not a native bacterial protein. Briefly describe your scheme.
Ans: The X gene is cloned into an appropriate plasmid through the use of restriction enzymes and ligase. The plasmid is then introduced into a bacterial host and the cells should produce protein X (as long as it is not toxic to the host).
Difficulty: 3

102. Design an experiment that utilizes suggestive physical evident to test the hypothesis that during recombination DNA molecules break and rejoin.
Ans: Bacteriophage previously grown in either heavy or light isotopes of nitrogen or carbon can be used to track breakage by infection of bacteria and subsequent CsCl gradient centrifugation. Banding at densities intermediate to the control heavy and light phage DNA indicates the occurrence of breakage and rejoining.
Difficulty: 4

103. Viruses may have RNA, ssDNA or dsDNA as their molecule of heredity. Design an experiment that would allow you to test an unknown virus and determine which it carried.

Ans: Grow bacteria infected with virus in the presence of radioactive uracil. Isolated genetic material from phage will be radioactive if they are an RNA containing virus. Double stranded versus single stranded DNA can be resolved by the ability to undergo digestion with restriction enzymes.

Difficulty: 4